
What does self-supervision add to engineered features for stellar light curves?

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Abstract

Self-supervised models promise one representation, pre-trained once on unlabelled survey data and read out cheaply for many downstream tasks. In stellar time-domain astronomy that promise competes with a strong, cheap incumbent: a few dozen hand-engineered light-curve statistics. We train a Conv1D variational encoder with a recurrent latent-dynamics model on 195,883 unlabelled TESS two-minute light curves, freeze it, and evaluate the resulting star embedding on ten labelled tasks spanning variability classification, asteroseismology and rotation, against three baselines: the engineered features alone, an architecture-matched supervised Conv1D trained end-to-end, and an MLP on raw flux. Concatenating the learned embedding with the engineered features improves a frozen linear readout beyond 2 SE on 7 of 10 tasks, while an untrained encoder does not. The same frozen pipeline is ahead of the supervised Conv1D on 6 of 10 tasks and indistinguishable from it on the other 4, and ahead of the raw-flux MLP on 9 of 10. **A learned representation and hand-built features are complementary rather than competing, and once combined, plain logistic regression on frozen features matches what end-to-end deep networks reach on the labels that exist.**

1 Introduction

Wide-field photometric surveys now produce light curves far faster than they can be labelled. TESS [Ricker et al., 2015] has observed hundreds of thousands of bright stars at two-minute cadence, and LSST [Ivezić et al., 2019] will enlarge the gap by orders of magnitude. Self-supervised learning is the natural response: pre-train one encoder on the unlabelled archive, then attach a cheap readout for each scientific question. This shape has recently been pursued for astronomical light curves by Donoso-Oliva et al. [2025], Zuo et al. [2025] and Rui [2026], and for TESS by Poznanski [2026], who validates with embedding diagnostics rather than downstream tasks.

What makes this domain unusual is the strength of the incumbent it must improve on. Decades of variable-star research have distilled light curves into compact engineered summaries – periodogram peak structure, band-limited power, spectral entropy, robust scatter statistics – and pipelines built on them remain highly competitive [Audenaert et al., 2021]. The benchmark of Li et al. [2025] makes the point sharply: across seven expert-labelled variable-star classes, no time-series foundation model they tested strictly surpassed the long-established domain feature baseline on classification. A learned representation for this data therefore has to answer a specific question, not a generic one: does it carry information the engineered features do not already have?

We take that question literally and measure it. We pre-train a latent-dynamics encoder on unlabelled TESS photometry, freeze it, and ask what its embedding adds to a 25-feature engineered stack under an identical linear readout – and whether the resulting pipeline, which trains no deep network

at readout time, is competitive with deep networks trained end-to-end on the same labels. Our contributions are:

- **The learned embedding is complementary to engineered features, not a replacement for them.** Concatenating the two improves a frozen linear readout beyond 2 SE on 7 of 10 tasks, with the gains concentrated where the engineered basis is information-starved; an untrained-encoder control shows the effect is not generic column addition.
- **The combined pipeline, read out by logistic regression, is competitive with end-to-end deep learning on these labels.** Against an architecture-matched supervised Conv1D it is ahead on 6 of 10 tasks and indistinguishable on the other 4; against an MLP on raw flux it is ahead on 9 of 10.

2 Data and method

Data. We use TESS two-minute SPOC PDCSAP photometry for 195,883 stars brighter than $T = 10$. Each light curve is split into contiguous segments at runs of missing cadences and at time gaps exceeding five times the median cadence; segments are median-normalised by their median absolute deviation and sliced into non-overlapping windows of 256 cadences (≈ 8.5 h). Missing values are never interpolated or padded – any window containing one is discarded, and segments are never stitched across sector or gap boundaries. Pre-training consumes sequences of 16 consecutive windows (≈ 5.69 d) and uses no labels of any kind.

Encoder. A 1D convolutional variational encoder [Kingma and Welling, 2014] maps each window to a 128-dimensional Gaussian posterior, and a GRU [Cho et al., 2014] predicts the next window’s latent from its predecessors, run both forward and backward over the sequence. The training objective combines window reconstruction, the KL term, the latent-dynamics prediction error, and a Hann-tapered spectral auxiliary term that penalises reconstruction error in the log periodogram. The encoder is trained once with six random seeds and then frozen; no gradient from any label ever reaches it.

Representations and readout. A star is represented by μ , the mean over its first segment (16–20 windows) of the per-window posterior means. The engineered baseline is a fixed 25-feature vector modelled on the T’DA variability pipeline [Audenaert et al., 2021]: the three strongest periodogram peaks with their normalised powers and ratios, a harmonic-power ratio, power in eight log-spaced frequency bands, spectral entropy and centroid, skew, kurtosis, point-to-point scatter ratio, and three robust dispersion statistics. Readouts are deterministic and frozen, standardised on the training split alone: class-balanced L_2 logistic regression at a fixed $C = 1$ for the detection tasks, ridge for the two regressions with its penalty set by leave-one-out cross-validation. Cross-validating C as well changes the reported gains by at most 0.011, and the two tasks that move by more than 0.008 both move in the fusion arm’s favour. We report four arms – features, μ , features $\oplus \mu$, and the same fusion built on an *untrained* encoder as a control.

Supervised baselines. Two end-to-end baselines see exactly the same input windows, star sets and splits. **C1** reuses the encoder’s convolutional trunk verbatim, adds a bottleneck and a scalar head, and is trained per task with a star-level bag loss (the mean of the star’s window logits, taken in-graph) and class weighting. **C2** replaces the convolutional trunk with a dense one and changes nothing else, so C1 versus C2 isolates the convolutional inductive bias. Both use one pre-registered regularisation setting for all tasks, select on a validation split no probe has ever touched, and are trained with three seeds.

Tasks and metrics. We score ten tasks with one probe per physical quantity: pulsating variables [Gao et al., 2025], eclipsing binaries [Prša et al., 2022], rotation and rotation period [Boyle et al., 2026], transits [Guerrero et al., 2021], flare activity [Seli et al., 2025, Vida et al., 2021], oscillating red giants and their $\nu_{\max}$ [Hon et al., 2021], solar-like oscillators [Hatt et al., 2023], and RGB versus helium-burning discrimination [Sreenivas et al., 2026]. Here $\nu_{\max}$ is the frequency of maximum oscillation power. Detection tasks are scored by the area under the precision–recall curve (PR-AUC), whose floor is the positive-class prevalence rather than 0.5; prevalence is therefore quoted with every detection score. The two regressions are scored by R^2 .

Table 1: Ten-task scorecard. Linear-readout arms use a frozen logistic or ridge readout on the mean-pooled representation; the two supervised arms are trained end-to-end per task. Bold marks a leader separated from the runner-up by the combined 2 SE band; italic marks a leader that is only the point-estimate armax. n is the test-star count and prev. the positive-class prevalence.

task	metric	n (prev.)	frozen linear readout			end-to-end supervised	
			features	μ	features $\oplus \mu$	Conv1D	MLP
pulsating	PR-AUC	2021 (0.107)	0.789	0.806	0.846	0.816	0.748
eclipsing binary	PR-AUC	2021 (0.097)	0.742	0.771	<i>0.775</i>	0.774	0.719
rotation	PR-AUC	2021 (0.089)	0.540	0.559	<i>0.555</i>	<i>0.565</i>	0.526
transit	PR-AUC	2021 (0.060)	0.190	0.144	0.218	0.134	0.109
oscillating giant	PR-AUC	3429 (0.383)	0.920	0.854	0.918	0.875	0.883
solar-like osc.	PR-AUC	3429 (0.122)	0.320	0.336	0.390	<i>0.396</i>	0.275
RGB vs. HeB	ROC-AUC	161 (0.702)	0.758	0.660	0.708	0.530	0.762
$\nu_{\max}$	R^2	1313	0.831	0.802	0.867	0.832	0.834
rotation period	R^2	150	0.703	0.677	0.717	0.597	0.698
flare activity	PR-AUC	3429 (0.089)	0.474	0.453	0.524	<i>0.533</i>	0.444

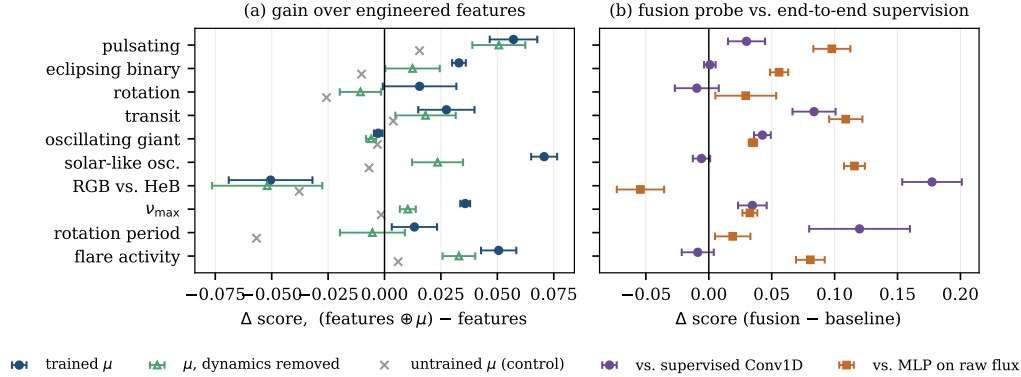


Figure 1: The two claims. **(a)** What the learned embedding adds to the engineered features under an identical frozen linear readout, for the shipped encoder, the same recipe with the latent-dynamics term removed, and an untrained-encoder control. **(b)** The frozen fusion probe against the two end-to-end supervised baselines. Bars are 2 SE; the untrained control is a single encoder and carries none.

Uncertainties. All error bars are $2\times$ the standard error of the mean over seeds. For the μ arms the seeds are independent pre-training runs (six); for C1 and C2 they are initialisation and shuffling seeds (six, matching the pre-training count). Because a pre-training seed and an initialisation seed are different objects, differences between a μ arm and a supervised arm are treated as unpaired and their bands added in quadrature even at equal counts. The engineered arm is deterministic and carries no seed spread. A difference counts as resolved only when it exceeds the combined 2 SE band.

3 Results

The learned embedding adds to the engineered features. Table 1 and Figure 1a give the comparison. Concatenating μ with the 25 features improves the frozen readout beyond 2 SE on 7 of 10 tasks, led by solar-like oscillators $+0.071$, pulsating $+0.057$ and flare activity $+0.051$, and reaching $+0.013$ (R^2) on rotation period. Rotation is positive but just short of the bar ($+0.016$ against 0.016). The two negative tasks are the ones where the engineered basis is already saturated or the probe is small: oscillating giants (-0.003 , on a baseline of 0.920) and RGB versus helium-burning (-0.051 , 161 test stars). On both, an untrained-encoder control loses by an amount we cannot resolve from the trained fusion's – the signature of extra collinear columns diluting a linear fit, not of μ carrying misleading content; Appendix A gives both rows in full. Across all ten tasks the untrained control is negative on seven and never exceeds $+0.016$, so the gains are specific to what pre-training learned. μ

on its own does not replace the features: it is ahead on only 4 of 10 tasks alone, which is the same conclusion Li et al. [2025] reach for foundation-model embeddings, and is why we report fusion rather than substitution [Makinen et al., 2024].

The frozen pipeline is competitive with end-to-end deep learning. Against C1 – the same convolutional trunk, the same labels, the same input windows, trained end-to-end – the frozen fusion probe is ahead on 6 of 10 tasks and indistinguishable from it on the other 4 (Figure 1b). Against C2, an MLP on raw flux, it is ahead on 9 of 10 and behind on one (RGB versus helium-burning, 161 stars). C1 is not weak: it clears the engineered arm on 5 of 10 and the μ -only arm on 4, and on eight tasks the frozen probe recovers 0.98–1.09 of its margin above the metric floor. The correct reading of this comparison is *not* that self-supervision matches supervision in general. It matches what this architecture reaches on the labels that actually exist, which range from 669 to 16,002 labelled stars per task.

The latent-dynamics term is what buys the complementarity. Replacing the forward–backward dynamics objective with a dynamics-free variant, all else fixed and the same six seeds, reduces the fusion gain on all 10 tasks, and by more than 2 SE on 6 when the two arms are differenced seed by seed (Figure 1a, open triangles; largest losses solar-like oscillators -0.047 , rotation -0.026 , $\nu_{\max}$ -0.025). The recurrent prediction term, not the autoencoder alone, supplies the information the engineered features lack.

4 Discussion

Selecting an encoder is harder than training one. Our largest practical difficulty was not optimisation but model selection. The reconstruction objective develops a localised artifact at the window boundary, and none of eleven interventions across four families of fix satisfied both pre-registered acceptance criteria, so the tapered variant shipped on the stated rule, not because it won. Underlying this is a measurement we think generalises. Across ten pre-training recipes, validation reconstruction *anti*-predicts downstream probe score: the Spearman ρ between negated validation reconstruction and PR-AUC runs from -0.58 to -0.84 over four tasks, the loss-optimal recipe is the dynamics-free one on all four, and trusting the loss costs 0.021–0.049 PR-AUC. The effect lives across recipes, not across seeds of one recipe, where no label-free metric we tried ranks anything. Self-supervised representation work in this domain therefore cannot be tuned on its own objective and must be gated on a held-out probe.

Limitations. Every comparison here is made under a linear readout, which is a choice rather than a concession: logistic regression over named axes is cheap, auditable, and stable enough that a change in the data appears in the coefficients rather than in the optimiser. We claim complementarity in that setting, not for readouts of arbitrary capacity. Our corpus is the bright ($T < 10$) two-minute sample, which is better studied and less confused than the faint population a survey-scale claim would need. Every arm sees only each star’s first segment (≈ 5.7 d), which equalises access but bounds the supervised baselines at the probe’s input scope rather than in general. Our labels come from incomplete catalogues, so a model-positive, catalogue-negative star scores as a false positive even when genuine. Rotation-period regression saturates at the 5.69-day input baseline – stars with true periods averaging 8.60 and 17.34 days are both answered near 5.6 – so its R^2 is not period inference beyond the segment length. Flare is an activity label, not a detection label: 81% of labelled positives show no catalogued flare inside the encoded segment, so we claim flare propensity only; auditing its negative class against the detector’s own published search cut moved the gain from $+0.0506$ to $+0.0507$. Under this frozen readout the advantage is also a large-label phenomenon, narrowing on 6 of 10 tasks as the labelled budget falls towards one percent.

Reproducibility and disclosure. §2 gives the corpus cut, geometry, features, objective and readouts; the remaining sizes are four convolutional stages of width 32/64/128/256, kernel 5, each halving length, a 128-dimensional Gaussian latent, and a single-layer GRU of width 256 across the 16 windows. Splits are by star, drawn once, and shared byte-for-byte by every arm, including both supervised baselines, which reproduce the probe’s star sets and test counts exactly. Code and score tables will be released on publication; generative AI assisted with analysis code and prose editing.

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206 A The two tasks where fusion does not help

task	features	fusion Δ	2 SE	untrained Δ	n_{test}
oscillating giants	0.920	−0.003	0.002	−0.003	3,429
RGB vs. He-burning	0.758	−0.051	0.019	−0.038	161

208 **Oscillating giants: the engineered basis is already near-sufficient.** The labels of Hon et al. [2021]
 209 come from a convolutional classifier reading a 128×128 image of each star’s power spectrum, which
 210 fires on a visible oscillation power excess at $6 \lesssim \nu_{\text{max}} \lesssim 250 \mu\text{Hz}$. Our engineered vector encodes
 211 close to that same quantity: eight log-spaced band powers spanning 0.3–360 c/d, plus the spectral
 212 centroid and entropy, locate the excess and describe its concentration. The features accordingly reach
 213 0.920, and the trained and untrained fusion arms then lose the same 0.003 – 128 collinear columns
 214 diluting a linear fit, with no trained-versus-untrained difference left to attribute to μ .

215 **RGB versus helium-burning: no arm resolves the discriminant.** Sreenivas et al. [2026] assign the
 216 evolutionary state with a convolutional classifier reading the background-corrected power spectrum
 217 *folded at $\Delta\nu$* ; the physics underneath is the dipole mixed-mode period spacing $\Delta\Pi_1$ [Bedding
 218 et al., 2011]. Neither is accessible at our baseline. The encoder sees one 256-cadence window at a
 219 time, so the finest frequency structure available to μ is $\Delta f = 32.6 \mu\text{Hz}$, while $\Delta\nu$ for these stars
 220 is $\approx 1.5\text{--}18 \mu\text{Hz}$ – the comb itself is unresolved, let alone a fold of it. The engineered features,
 221 computed on the whole 5.69 d segment, reach $2.0 \mu\text{Hz}$, still coarser than the mixed-mode spacing
 222 $\Delta\Pi_1\nu^2 \approx 0.2\text{--}0.6 \mu\text{Hz}$ near $\nu_{\text{max}} = 50 \mu\text{Hz}$. No arm on this row measures the quantity the label
 223 encodes; the 0.758 the features reach rides on correlates of evolutionary state rather than on its
 224 discriminant. The row behaves accordingly: it is the only one where C2 (0.762) is ahead of the frozen
 225 probe while C1 reaches only 0.530 on 755 training stars, and it carries the smallest test set in the
 226 study.